MUKESH KUMAR S

Biomedical Engineer | Medical Device R&D | Healthcare AI & Medical Imaging

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Biomedical Instrumentation | Embedded Systems | Medical Imaging | Wearable Health Technology

PROFESSIONAL SUMMARY

Biomedical Engineering graduate focused on Medical Device R&D and healthcare technology, with hands-on experience in biomedical sensor integration, embedded systems, wearable monitoring, medical imaging, healthcare AI, and MATLAB-based biomechanical simulation. Experienced with MAX30205, MAX30102, ESP32, MEMS microphones, BLE, deep learning, and medical image analysis, with knowledge of ISO 13485, ISO 14971, and IEC 60601-1.

TECHNICAL SKILLS

Programming: C, C++, MATLAB
Biomedical Instrumentation: Biomedical sensors, physiological measurement, medical equipment, sensor calibration
Embedded Systems: Arduino, ESP32, Arduino firmware, biomedical sensor integration, BLE
Biomedical Sensors: MAX30205, MAX30102, MEMS microphone
AI & Medical Imaging: Deep learning, multilabel classification, medical image analysis, chest X-ray analysis, healthcare AI
Signal Processing: Biomedical signal processing, Signal Processing Onramp
Medical Device Development: Prototype development, testing, calibration, technical documentation
Medical Device Standards: ISO 13485, ISO 14971, IEC 60601-1
Tools: MATLAB, Arduino IDE, MS Excel

PROJECTS

AI-Based Multi-Modal Neonatal Health Assessment System *2026*

- Engineered a proposed wearable neonatal monitoring system integrating baby cry/audio, body temperature, SpO_2, and heart-rate acquisition.
- Integrated MEMS microphone, MAX30205, MAX30102, and ESP32 with BLE-based data transmission for remote monitoring.
- Proposed AI-enabled multimodal health analysis with cloud/mobile dashboard integration and defined accuracy, battery life, comfort, and clinical usability as key validation requirements.

Multilabel Classification of Lung Diseases Using Deep Learning *2026*

- Developed a deep-learning approach for multilabel classification of abnormalities from chest X-ray medical images.
- Investigated classification of pneumonia, tuberculosis, and COVID-related abnormalities for computer-aided medical image analysis.

Smart Biomedical Thermometer Using MAX30205 *2026*

- Developed a sensor-based digital thermometer using MAX30205 and Arduino for human body-temperature measurement.
- Implemented firmware-based temperature acquisition, monitoring, and calibration procedures.

Simulation of 3D Knee Contact Model Using MATLAB *2025–2026*

- Developed a MATLAB-based 3D knee joint contact simulation for biomechanical analysis of knee loading and contact behavior.

EDUCATION

B.E. Biomedical Engineering *2022–2026*
E.G.S. Pillay Engineering College, Nagapattinam **CGPA: 8.03/10**

Higher Secondary – Bio-Maths *2021–2022*
Laurel Higher Secondary School **75%**

INDUSTRIAL & HOSPITAL EXPOSURE

Guru Multispecialty Hospital, Madurai – Biomedical equipment, medical instruments, and clinical workflow exposure *May 2025*

Dr. Rao Hospital, Madurai – Medical instruments and biomedical equipment exposure *May 2025*

ITI, Madurai – Industrial and technical exposure *2025*

CERTIFICATIONS & TRAINING

- Signal Processing Onramp – MathWorks
- Advanced Critical Care Equipment Training – KAV Biomedical Equipment Training and Education
- Advanced-Level Calibration and Handling Training on Critical Care Equipment – KAV Biomedical Equipment Training and Education
- Integration of Clinical Knowledge and Medical Coding in Healthcare System – Theeram Healthcare, Madurai
- Microbial Biotechnology – 3-credit NPTEL Course, SWAYAM, Jan–Apr 2025

ACHIEVEMENTS

- Represented the Biomedical Engineering Department at Science Expo, E.G.S. Pillay Engineering College, August 2024.
- Recognized for active participation in Science Expo 23.
- Participated in technical sessions on Biomedical Instrument Design and Generative AI.
- Participated in an inter-college quiz competition organized by Shankar IAS Academy.